GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2023-24) -- SEMESTER 6

COURSE: COMPILER DESIGN (ETCS-602)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: ETCS-602

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) List all 6 phases of a compiler and their input/output artifacts.

Q1. (b) What is difference between a token, a pattern, and a lexeme?

Q1. (c) What is left recursion in CFG? Write formula to eliminate immediate left recursion.

Q1. (d) What is left factoring in context-free grammars?

Q1. (e) Define handle pruning in shift-reduce bottom-up parsing.

Q1. (f) What is difference between S-attributed and L-attributed SDD definitions?

Q1. (g) What is Three-Address Code (TAC)? List quadruples, triples, and indirect triples.

Q1. (h) What is basic block and flow graph in code optimization?

Q1. (i) Explain loop optimization techniques: Code motion, Induction variable elimination, Loop unrolling.

Q1. (j) What is DAG (Directed Acyclic Graph) in compiler optimization?

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) For grammar E -> E + T | T, T -> T * F | F, F -> (E) | id, compute FIRST and FOLLOW sets and construct LL(1) parsing table.

Q2. (b) Construct Canonical LR(0) collection of items and parse table for grammar S -> C C, C -> c C | d.

Q2. (c) Construct LALR(1) parsing table for grammar S -> A a | b A c | B c | b B a, A -> d, B -> d.

Q2. (d) Generate Three-Address Code, Quadruple, and Triple representations for: x = (-a * b) + (c - d) / (e + f).

Q2. (e) Construct DAG for basic block: a = b + c; d = a - b; e = d + c; f = e + a; and perform common subexpression elimination.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Explain the role of Lex / Flex lexical analyzer generator with transition diagrams. [10 Marks]

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(b) Explain buffer pairs and sentinel techniques in lexical analysis scanning. [10 Marks]

Q4. (a) Explain Operator Precedence Parsing with leading and trailing relations table. [10 Marks]

-- OR --

(b) Compare LR(0), SLR(1), CLR(1), and LALR(1) parsers based on states, lookaheads, and power. [10 Marks]

Q5. (a) Translate switch-case and while-loop control constructs into Three-Address Code with backpatching. [10 Marks]

-- OR --

(b) Explain type expressions, type equivalence, and type checking system for programming languages. [10 Marks]

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Q6. (a) Explain Principal Sources of Code Optimization: Constant folding, dead code elimination, copy propagation. [10

-- OR --

(b) Explain Dataflow Analysis equations for reaching definitions, available expressions, and live variables. [10 Marks]

Q7. (a) Explain register allocation and assignment using graph coloring algorithm. [10 Marks]

-- OR --

(b) Explain peephole optimization techniques: Redundant load/store elimination, unreachable code, algebraic simplification. [10 Ma

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****